A data-driven fake factor, learned event by event

SLT input production, preprocessing and first closure checks

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$HH \to bb\tau\tau$, lep-had (SLT)

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The training foundation is now in place

We want a smooth, event-level fake factor while keeping the central estimate anchored to **data**.

✓ **Inputs**
ID and anti-ID H5 files

✓ **Preprocessing**
labels, weights and folds

✓ **Validation**
binned FF and MLP closure

The inputs reproduce the established method and are ready for SALT training.

The central fake rate remains a data measurement

A jet faking a τ_{had} is the main reducible background. The fake factor is measured from **true-τ-subtracted data**:

$$\mathrm{FF} = \frac{N_{\text{ID}}^{\text{data}-\text{true-}\tau}}{N_{\text{anti-ID}}^{\text{data}-\text{true-}\tau}},$$

Measure

ID and anti-ID data, after subtracting genuine τ contamination.

Apply

Weight anti-ID data by FF to predict the fake yield in the ID region.

Today FF is binned in p_T^{τ} and split by prong and η; the ML model replaces only this low-dimensional parameterisation.

Classifier odds provide the event-level fake factor

We keep the physics of the method — **the rate is measured in data** — but replace the binned histogram with a smooth, high-dimensional function.

- Train a classifier to separate **ID vs. anti-ID** candidates on the same true-τ-subtracted data.
- With $p(x) = P(\text{pass } \tau\text{-ID} \mid x)$, the *odds* are the per-candidate fake factor:

$$\mathrm{FF}(x) = \frac{p(x)}{1 - p(x)}.$$

- Simulation is used only for the true-τ subtraction and for cross-checks — never to set the central rate.

Same data-measured quantity; more information, smooth interpolation and no bin edges.

Both training classes are present in the H5 inputs

We built the training inputs in **both** regions, keeping **data** + **true-τ simulation** and excluding signal (a fake template needs no signal).

- 43 physics processes, each in ID and anti-ID
- $\sim$7.5M candidate events, all files healthy
- signal correctly absent by construction

region	data candidates
ID (pass τ-ID)	133,484
anti-ID	732,107

Both regions contain the subtraction information and the full kinematics needed by SALT.

Signed weights build the two data-driven fake classes

Each class represents data − true-τ simulation, implemented as a **per-candidate weight** rather than a cut.

1. **Label** = region: ID $\rightarrow$ 1, anti-ID $\rightarrow$ 0.
2. **Weight**: data enters positively; truth-matched τ's in simulation enter negatively.
3. **Fold** by event number: three train, one validation, one test.

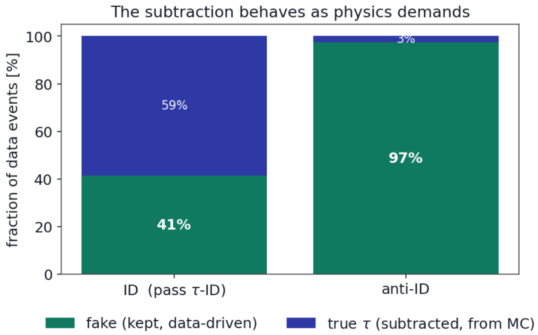
The per-candidate weight:

$$w = \begin{cases} +1 & \text{data} \\ -w_{\mathrm{MC}} & \text{true-}\tau\text{ simulation} \\ 0 & \text{other simulation} \end{cases}$$

Each class is therefore the data-driven fake component.

6.6M candidates enter five folds; truth sets only the weight, and no candidate is removed.

Anti-ID is 97% fake, while ID requires a large subtraction



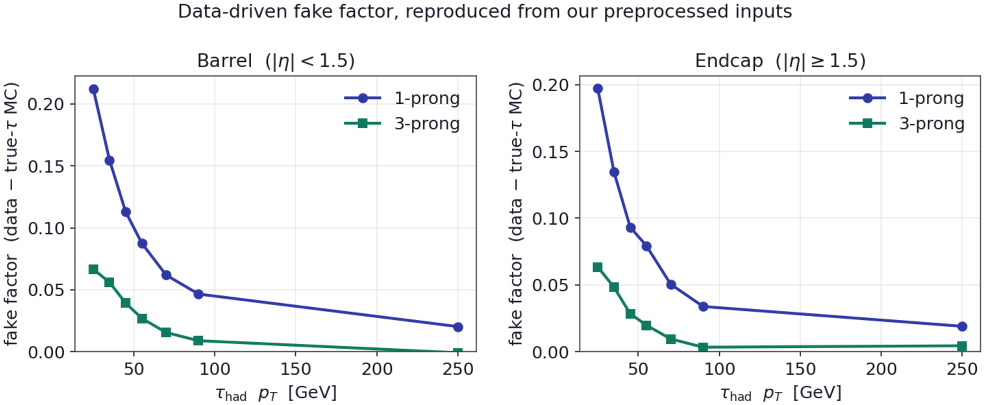
- In **anti-ID**, 97% of data is fake — a clean, fake-enriched sample, as it should be.
- In **ID**, more than half is genuine τ; the subtraction removes it, leaving the fakes.
- The inclusive data-driven fake factor is

$$\mathrm{FF} \approx 0.08,$$

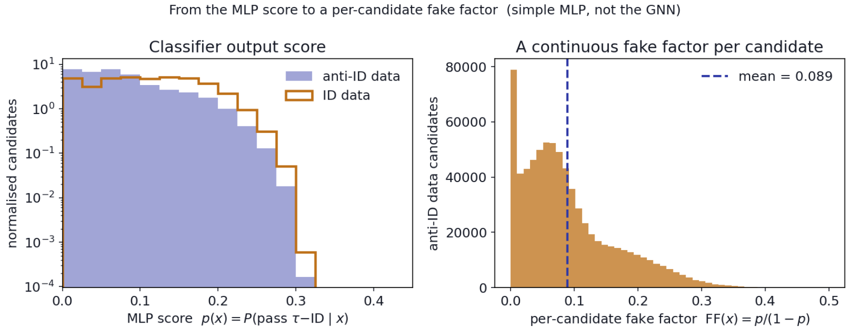
right in the expected range.

The preprocessed inputs reproduce the expected FF shape

The fake factor falls with p_T^τ; 1-prong candidates remain above 3-prong in barrel and endcap.



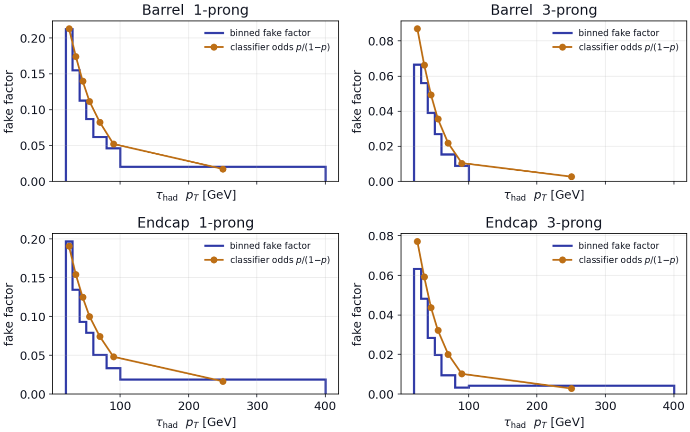
From the MLP score to a per-candidate fake factor



ID candidates score higher; the odds $p/(1-p)$ convert each score into a smooth fake factor with no bin edges.

A simple MLP already closes against the binned result

Preview with a simple MLP (NOT the GNN): the odds already track the fake factor
(same data − true-τ subtraction; smooth where the binned method has one wide bin)



We are ready to move from preprocessing to SALT

Completed

- ✓ Inputs produced in both regions, data + true-τ simulation, no signal.
- ✓ Preprocessing done: subtraction as a weight, tails preserved, 5 folds.
- ✓ Validated: reproduces the established fake factor; odds match it.

Next

- ▶ train the fake-factor network (5-fold), then close it against the binned measurement on the trained output;
- ▶ benchmark against the current method in the signal and validation regions.

Immediate milestone: five-fold SALT training and closure in signal and validation regions.

Backup: why the odds are the fake factor

Train on data to separate class $1 =$ ID from class $0 =$ anti-ID. The classifier learns $p(x) = P(1 \mid x)$. By Bayes, with class densities f_1, f_0 and training proportions π_1, π_0:

$$p(x) = \frac{\pi_1 f_1(x)}{\pi_1 f_1(x) + \pi_0 f_0(x)} \implies \frac{p(x)}{1 - p(x)} = \frac{\pi_1}{\pi_0} \frac{f_1(x)}{f_0(x)}.$$

The per-candidate fake factor is the local ID/anti-ID ratio:

$$\mathrm{FF}(x) = \frac{N_1(x)}{N_0(x)} = \frac{p(x)}{1 - p(x)} \cdot \underbrace{\frac{N_1/N_0}{\pi_1/\pi_0}}_{c}.$$

- We train with **natural proportions**, so $c = 1$ and $\mathrm{FF}(x) = p/(1 - p)$ directly.
- The true-τ subtraction enters as a per-candidate weight ($+$ for data, $-$ for true-τ simulation), so each class is the data-driven fake component.

Backup: training inputs at a glance

Inputs

- 43 processes, ID and anti-ID, data + true-τ simulation, no signal
- $\sim$7.5M candidates; data: 133k (ID) / 732k (anti-ID)

Preprocessing

- target: ID $= 1$ vs anti-ID $= 0$
- weight: $+$data, $-$true-τ; fakes ($p/(1-p)$) kept, nothing dropped
- 6.6M candidates, 5 folds

What the classifier sees

- τ candidate: p_T, η, prong, plus the full object list
- event topology: m_{bb}, p_T^{HH}, $\Delta R_{\tau\ell}$, MET, H_T, jet kinematics, ...
- *not* used: truth information (only for the subtraction weight)

A note for training

- signed weights need care — we monitor stability and can switch to a positive-weight subtraction if needed.